

- Shunt sampling : - single node / parallel structure
 $R_{eff} = \frac{R_{out}}{1 + \beta A_{ol}}$

↓
 If we short output
 we kill feedback,
 (if we cut output nothing happens)

- Series sampling : - single loop / series structure
 $R_{eff} = R_{out} (1 + \beta A_{ol})$

↓
 If we cut output,
 we kill feedback
 (if we short output nothing happens)

→ to be measured with V_{in} in one point of the single loop, usually $R_{out} = \infty$ because the dependent generator is turned down.

IV 2021

Ex 1

$$V_G = \frac{V_{DD}}{2}$$

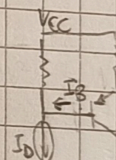
$$V_{GS} = V_G - I_D R_S \rightarrow I_D = \left(\frac{V_{DD}}{2} - V_{GS} \right) \frac{1}{R_S} = I_{DSS} \left(1 - \frac{V_{GS}}{V_T} \right)^2$$

$$\left(\frac{V_{DD}}{2} - V_{GS} \right) \frac{1}{R_S I_{DSS}} = \left(1 - \frac{V_{GS}}{V_T} \right)^2 \rightarrow V_{GS1} = -2.16 V$$

$$V_{GS2} = 6.162 V > V_T \quad \text{OK}$$

$$\left(\frac{I_D - I_C}{\beta F} \right) R_D = I_C R_E \left(1 + \frac{1}{\beta F} \right) - V_{BE}$$

$$I_C = \frac{I_D R_D + V_{BE}}{\frac{R_D}{\beta F} + R_E \left(1 + \frac{1}{\beta F} \right)} = 0.524 \text{ mA}$$

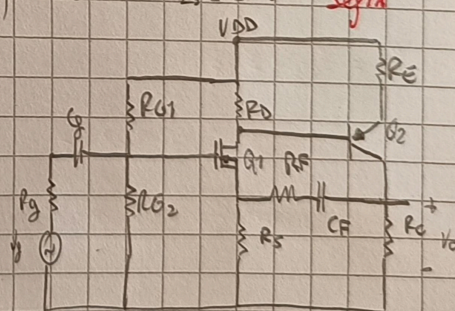


$$V_{DS} = V_{DD} - R_D \left(I_D - \frac{I_C}{\beta F} \right) - I_D R_S = 6.8 V$$

$$V_{CE} = - \left(V_{DD} + I_C R_E \left(1 + \frac{1}{\beta F} \right) - I_C R_C \right) = -I_C R_C + I_C R_E \left(1 + \frac{1}{\beta F} \right) - V_{DD} = -2.59 V \quad \text{separi}$$

$$Q_1(I_D, V_{DS}) = (0.338 \text{ mA}, 6.8 V)$$

$$Q_2(I_C, V_{CE}) = (0.524 \text{ mA}, -2.59 V)$$



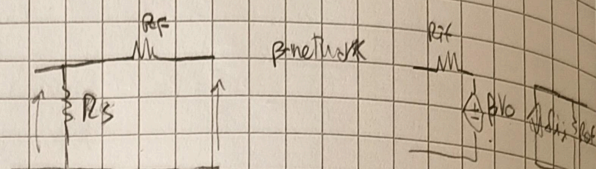
with caps shorted: voltage mixing, voltage sampling, voltage amplifier

$$A_v = \frac{V_o}{V_s} = ?, R_{in} = ?$$

Trueman, single loop

$$R_G' = R_G \parallel R_{G2} = 50 \text{ k}\Omega$$

$$V_g' = V_g \frac{R_G'}{R_g + R_G'} = \frac{0.98}{0.98}$$



$$R_{if} = R_S \parallel R_F = 952 \Omega$$

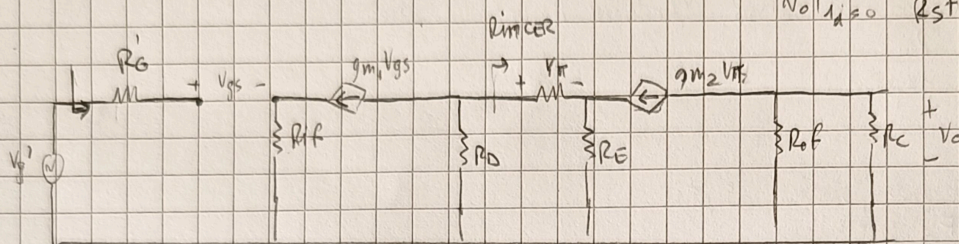
$$R_{of} = R_F + R_S = 21 \text{ k}\Omega$$

$$\beta = \frac{V_o}{V_i} \bigg|_{i_d=0} = \frac{R_S}{R_S + R_F} = 0.0476$$

$$g_{m1} = \frac{2I_D}{V_{GS} - V_T} = 0.58 \text{ mS}$$

$$g_{m2} = \frac{I_D}{V_T} = 0.62 \text{ mS}$$

$$r_{\pi} = \frac{\beta_{DQ}}{g_{m2}} = 4771 \Omega$$



$$A_v = \frac{V_o}{V_g'} \bigg|_{\beta \rightarrow \infty} = \alpha_{CS} \cdot \alpha_{CER}$$

$$R_{inCER} = r_{\pi} + (1 + \beta_{DQ}) R_E = 226.97 \text{ k}\Omega$$

$$\alpha_{CER} = \frac{-\beta_{DQ} (R_{of} \parallel R_L)}{R_{inCER}} = -2.985 \text{ [V/V]}$$

$$\alpha_{CS} = \frac{-g_{m1} (R_D \parallel R_{inCER})}{1 + g_{m1} R_F} = -2.067 \text{ [V/V]}$$

$$A_v = 6.11 \text{ [V/V]}$$

$$A_F = \frac{A_v}{1 + \beta_{DQ} A_v} = \frac{V_o}{V_g'} = 4.73 \text{ [V]}$$

$$A_v = \frac{V_o}{V_g} = A_F \cdot \frac{R_G'}{R_g + R_G'} = 4.639 \text{ [V/V]}$$

$$R_{in}^d = \infty$$

$$R_{in}^F = R_{in}^d (1 + \beta_{DQ} \alpha) = \infty$$

$$R_{in}^F = (R_{in}^F - R_G') \parallel (R_G \parallel R_{G2})$$

$$R_{in} = R_G \parallel R_{G2} = 50 \text{ k}\Omega$$

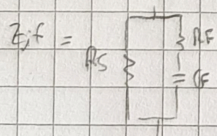
we calculate immediately because it is not affected by feedback

$$f_L = \frac{1}{2\pi g_{m1} R_{gSC}} = 312 \text{ Hz}$$

$$R_{gSC} = R_g + R_{in} = 51 \text{ k}\Omega$$

$$\boxed{C_g = 10 \text{ nF}} \\ \boxed{C_F = 1 \mu\text{F}}$$

considering capacitors of CF: we consider it in the beta network



$$R_{F1}^{SC} = R_F + R_S \parallel \frac{1}{g_{m1}} = 20.632 \text{ k}\Omega$$

$$R_{F2}^{SC} = R_F + R_S + R_L = 31 \text{ k}\Omega$$

$$p_2 = \frac{1}{2\pi} \left(\frac{1}{C_F R_{F1}^{SC}} + \frac{1}{C_F R_{F2}^{SC}} \right) \\ = \frac{1}{2\pi} \left(\frac{1}{C_F} \left(\frac{1}{R_{F1}^{SC}} + \frac{1}{R_{F2}^{SC}} \right) \right) \\ = 12.85 \text{ Hz}$$

$$p_2 = f_{Lg} + f_{LCF}$$

$$= 312 + 9.95 = 322 \text{ Hz}$$

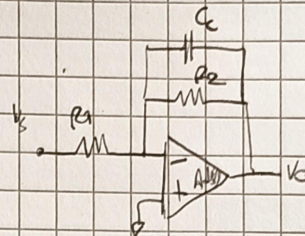
$$\text{contribution of } C_F \text{ to } f_L \leftarrow f_2^{CL} = f_{Lg} = 3.95 \text{ Hz}$$

we make an error of 3% by considering only C_g

Ex 2 $R_1 = 1k\Omega$
 $R_2 = 33k\Omega$

$A(s) \approx \frac{A_{d0}}{(1 + \frac{s}{\omega_{p1}})(1 + \frac{s}{\omega_{p2}})}$

$A_{d0} = 10^5$
 $\omega_{p1} = 10^2 \text{ rad/s}$
 $\omega_{p2} = 5 \cdot 10^3 \text{ rad/s}$



1) CC open: $T = -\frac{x_r}{x_s} = \frac{A_d(s) R_1}{R_1 + R_2}$

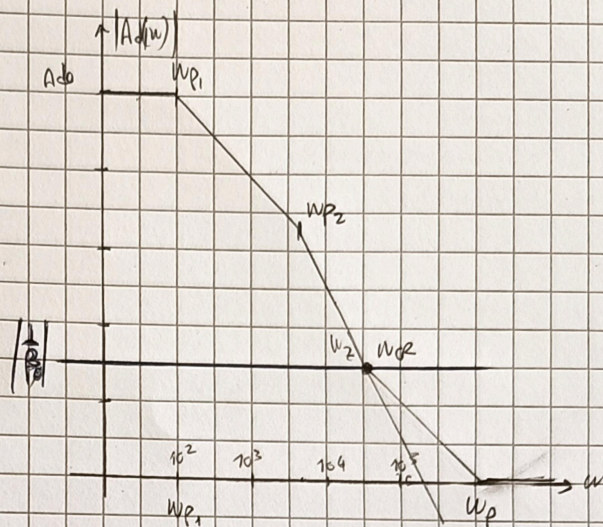
$\beta_o = \frac{1}{34}$, $\frac{1}{\beta_o} = \frac{R_1 + R_2}{R_1} = 1 + \frac{R_2}{R_1} = 34 = 30.6 \text{ dB}$

$\begin{cases} A_o \omega_{p1} = A_r \omega_{p2} \\ A_r \omega_{p2}^2 = \frac{1}{\beta_o} \omega_{cp}^2 \end{cases} \Rightarrow A_r = \frac{A_{d0} \omega_{p1}}{\omega_{p2}}$

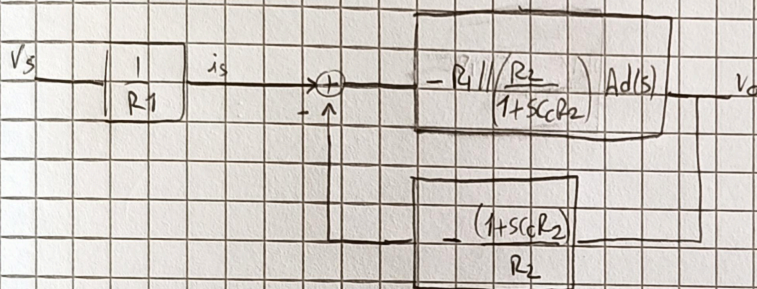
$\omega_{cp} = \sqrt{\beta_o A_{d0} \omega_{p1} \omega_{p2}} = 3.83 \cdot 10^4 \text{ rad/s}$

$PM = 180 - \angle \arg\left(\frac{\omega_{cp}}{\omega_{p1}}\right) - \angle \arg\left(\frac{\omega_{cp}}{\omega_{p2}}\right)$
 $= 7.58^\circ$ low phase margin

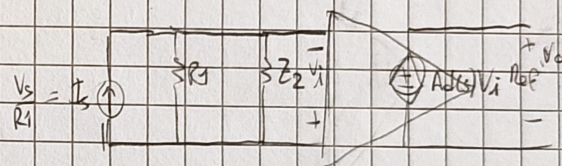
(dominant pole compensation can be applied because β_o is high enough)



2) current mixing - voltage sampling



$Z_2 = \left(\frac{1}{sC_c} \parallel R_2 \right) = \frac{R_2}{1 + sC_c R_2}$



$\beta_c = \frac{Z_1}{Z_1 + Z_2} = \frac{R_1}{R_1 + \frac{R_2}{1 + sC_c R_2}} = \frac{R_1}{R_1 + R_2} \frac{1 + sC_c R_2}{1 + sC_c \frac{R_2 R_1}{R_2 + R_1}} = \frac{R_1}{R_1 + R_2} \frac{1 + s/\omega_z}{1 + s/\omega_p}$

$\omega_z = \frac{1}{C_c R_2} = \omega_{cp} \rightarrow C_c = \frac{1}{R_2 \omega_{cp}} = 791 \text{ pF}$ $\gamma_{PM} = 0 \text{ dB}$

$\omega_p = \frac{R_2 + R_1}{C_c R_2 R_1} = 1.3 \cdot 10^6 \text{ rad/s}$

ω_{cp} is the same as before $\rightarrow PM_{comp} = PM - \angle \arg\left(\frac{\omega_{cp}}{\omega_p}\right) + \angle \arg\left(\frac{\omega_{cp}}{\omega_z}\right) = 50.9^\circ$

$\omega_{cp} =$

Tons